Code And Branch Coverage Report

Railway Reservation System — Backend

1. Introduction

Code coverage is a software testing metric used to measure how much of an application's source code is executed during automated testing. This report presents the statement/line, branch, and function coverage measured for the Railway Reservation System backend using c8 and Node.js's built-in test runner.

2. Objectives

• Measure the percentage of source statements executed during testing.
• Measure the percentage of decision branches executed during testing.
• Identify code areas that remain uncovered.
• Summarize the effectiveness and execution status of the existing backend tests.

3. Testing Environment

Component	Details
Operating System	Windows (test run from Windows command prompt)
Programming Language	JavaScript / Node.js
Application Framework	Express.js
Testing Framework	Node.js built-in test runner (node:test)
Coverage Tool	c8
Application	Railway Reservation System — Backend
Database / ODM	MongoDB / Mongoose

4. Coverage Measurement

Coverage was measured by running the backend test command through c8. The command used was:

```
npx c8 --reporter=text --reporter=html npm test
```

This command executes the test suite, prints coverage statistics in the terminal, and creates an HTML coverage report under the backend coverage directory.

5. Code Coverage Results

Metric	Result
Total statements	2,024
Covered statements	910
Missed statements	1,114
Statement / line coverage	44.96%

The measured statement/line coverage was 44.96% (910 of 2,024 statements/lines covered). The remaining 1,114 statements/lines were not exercised in this test run.

6. Branch Coverage Results

Metric	Result
Total branches	109
Covered branches	95
Missing branches	14
Branch coverage	87.15%

Branch coverage was 87.15% (95 of 109 branches covered). Fourteen branches remained uncovered according to the c8 summary.

7. Test Execution Summary

Metric	Result
Tests reported	37
Passed	37
Failed	0
Test execution time	4.60 seconds
Database integration tests	Not executed; MONGODB_TEST_URI was not configured
Function coverage	56% (14/25 functions)

The reported test run completed with 37 passing tests and no failures. Some database-dependent integration test groups did not execute because the MONGODB_TEST_URI environment variable was not configured. Therefore, the displayed coverage figures reflect the tests that ran and do not include those database integration scenarios.

8. Analysis

The backend achieved 44.96% statement/line coverage, 87.15% branch coverage, and 56% function coverage. Branch coverage is notably higher than statement coverage, indicating that the tests exercised a high proportion of measured decision branches while a substantial amount of code statements and functions remained unexecuted. The HTML report highlights uncovered source areas, including routes and services. Database-backed flows such as booking lifecycle, saved language preference, and waitlist monitoring were not covered in this run because the test database URI was unavailable.

9. Conclusion

Code and branch coverage analysis was successfully performed on the Railway Reservation System backend using c8 and the Node.js built-in test runner. The measured results were 44.96% statement/line coverage and 87.15% branch coverage, with 37 tests passing and zero failures. The results indicate that the current automated tests exercise a number of important validation and service behaviors, while additional tests—especially with a configured test database—can expand statement, function, and integration coverage.